ICARB-D ANNUAL MATURITY SURVEY

Dental Schools, Departments and Courses

Health informatics in dentistry: AI · Robotics · Bioinformatics · Software development

United States · national benchmark of dental schools

Annual questionnaire answered by the director of each course · version 3.0.0 · 2026-11-02

Main goal

Build a benchmark: one comparable number, from 0 to 100, that measures how mature each US dental school is in health informatics, with a score for each of its areas.

Proposed article title

A Benchmark of Health Informatics Maturity in US Dental Schools: A National, Annual, Course-Level Survey of Artificial Intelligence, Robotics, Bioinformatics and Software Development

Built on the PAHO/WHO–IDB AI Readiness Assessment Toolkit (2024), AI-GUARD (2026), Bias-free AI in Health (2026) and AI Prompt Design for Public Health (2025).

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1. About the study

Main goal: build a benchmark, one comparable number from 0 to 100, that measures how mature each US dental school is in health informatics, with a score for each of its areas. As a practical conclusion, the benchmark helps answer a funding question: if a grant for health informatics in dentistry were available, which university would have the best chance of using it well? Comparing schools, and comparing US schools with schools in other countries that answer the same survey, are secondary uses.

Health informatics is the umbrella. Its areas are health information systems (electronic records, imaging archives, coding, teledentistry, analytics), artificial intelligence, robotics and digital dentistry, bioinformatics, and software development for health. Every area depends on the same foundations: infrastructure, data and privacy, people, governance, and implementation and sustainability.

A dental school is organised as university → departments → courses. Each course director (the head of the course) answers one form for their course, because nobody knows the course better. Department scores are built from their courses and the university score from its departments, so the benchmark is grounded in what is actually taught and used. For each course the reader gets a complete, clear picture of what it has, who can do the work, what students learn and for how many hours, how technology is governed, and whether projects last.

The idea comes from the Pan American Health Organization (PAHO), which created a questionnaire to measure how ready public health institutions are to use AI: the Artificial Intelligence in Public Health: Readiness Assessment Toolkit (PAHO and the Inter-American Development Bank, 2024). This survey adapts its dimensions and 1–4 maturity scale to dental schools and extends them to the whole of health informatics. Governance items for AI tools follow PAHO's AI-GUARD tool (Yes / Partial / No), bias items follow PAHO's Bias-free AI in Health guidance, and generative-AI items follow PAHO's AI Prompt Design for Public Health.

Why now: AI already reads radiographs, digital workflows are replacing analogue ones, and students use generative AI every day, while schools invest at very different speeds and there is no shared number that says how ready each school is. The survey is repeated every year, so the benchmark also shows progress.

After answering, the same items work as a guide: each 1–4 item describes what level 4 (established) looks like, and the maturity checklist (section 4) lets a school see where it stands and which steps lead to the level it wants to reach.

Who answers

Questions the benchmark helps answer

- **Main goal.** Where does each US dental school stand in health informatics? The benchmark gives every school a number built from its departments and courses, with a score for each area.
- **Practical conclusion.** If a grant for health informatics in dentistry were available, which university would have the best chance of using it well?
- **Secondary.** How do schools compare with each other, and, where schools in other countries answer the same survey, how do US schools compare with them?

Health informatics and its areas

Health informatics is the umbrella and the benchmark measures it as a whole. Its areas are listed below; every area also depends on the same foundations.

Area of health informatics	What it covers
Health information systems	Electronic records, imaging archives, coding, teledentistry and analytics
Artificial intelligence	AI tools, AI teaching and generative AI
Robotics and digital dentistry	Robots, simulators, scanners, CAD/CAM and 3D printing

Area of health informatics	What it covers
Bioinformatics	Omics laboratories, pipelines and biological data
Software development for health	Building, testing and maintaining health software
Foundations	Infrastructure, data and privacy, people, governance, and implementation and sustainability

A dental school is organised as university → departments → courses. Each course director (the head of the course) answers one form for their course; department and university scores are built from the courses.

Level	How its score is obtained
University (dental school)	Score built from all its departments, weighted by the number of courses
Department	Score built from the courses it teaches
Course	One form, answered by the course director (the head of the course)

Objectives

- Build a 0–100 benchmark of health informatics maturity for every participating US dental school, with a score for each area (health information systems, AI, robotics and digital dentistry, bioinformatics, software development for health).
- Identify the schools whose foundations (infrastructure, data, people, governance, implementation) make them most likely to use new funding for health informatics well.
- Compare schools, and compare US schools with schools in other countries that use the same survey (secondary).
- Describe the physical resources available to each course: computing, AI tools, digital and robotic equipment, laboratories, datasets, connectivity and security.
- Describe human resources: faculty able to teach each domain, researchers, training received, technical staff and interdisciplinary partners.
- Quantify teaching: hours per year by topic for AI (including programming and model development), health information systems, robotics/digital dentistry and bioinformatics, at undergraduate, graduate and residency level.
- Assess governance, ethics, bias, generative-AI policy and regulatory alignment, using US references (HIPAA, FERPA, FDA, CODA, ADA standards).
- Assess implementation, evaluation and financial sustainability.
- Map each course's interest and capacity to contribute data, annotation and validation to a free, open dental AI model.
- Build a maturity index per domain for each course, department and school, summarise it by state and nationally, and track its change from year to year.

How the questionnaire is applied

- **Step 1.** Each year, the research team contacts the director of each participating course and completes Part A (course, department, university, survey year and consent) with them. Saving Part A on the study website generates a unique 6-character access code (format XXX-XXX). If the unit answered in a previous year, its previous code is recorded to link the waves.
- **Step 2.** The team sends the code and the website address to the respondent.
- **Step 3.** The respondent enters the code on the website (the only field on the login page) and answers Parts B–M. Answers are saved at every section change, so the form can be completed in several sessions. A public guide explains every question and what it captures.

- **Step 4.** After submission the form is locked. Each form produces one row in the data file (responses.csv), one column per item, ready for statistical analysis.

Respondent instructions

- Answer every item for the course named in Part A. For resources or rules provided by the department or school, describe what this course can use or must follow.
- Where an item asks about a course, write the course code and title.
- Each course has its own form and code; a department with several relevant courses sends one form per course.
- Use "Don't know / Not applicable" rather than guessing. Approximate numbers are acceptable.
- Hours are reported per academic term. Item B05 (terms per year) converts them into hours per year; programs with an annual system enter 1 term per year.

US context

Items marked "US reference" are read with the US laws and bodies below.

	United States
Degree	DDS/DMD
Curriculum / accreditation body	CODA (Commission on Dental Accreditation)
Professional and regulatory bodies	ADA, ADEA and state dental boards
Privacy law	HIPAA and FERPA
AI device regulator	FDA
Academic calendar	Programs use semesters, trimesters or quarters.
Languages	English

Readiness domains and scale

Maturity items use a 1–4 anchored scale adapted from the PAHO/WHO toolkit (1 = absent, 2 = initial, 3 = developing, 4 = established), plus "Don't know / Not applicable" (coded −1 and treated as missing). Each 1–4 item belongs to one of ten domains:

- **INF —** Physical infrastructure and technology
- **DAT —** Data management, quality and privacy
- **HUM —** Human resources and expertise
- **CUR —** AI curriculum and teaching
- **GEN —** Generative AI governance and literacy
- **HIN —** Health information systems
- **ROB —** Robotics and digital dentistry
- **BIO —** Bioinformatics and omics
- **SWD —** Software development for health
- **GOV —** Governance, ethics and regulation
- **IMP —** Implementation, evaluation and sustainability

2. Questionnaire

Part A. Profile of the course (completed by the research team)

Goal. Place each course in its department and university, and record who answers for it, so every score can be traced, compared and followed from year to year.

What this part captures. Places the course in the hierarchy university → department → course (with country, state and training level), records who answers and their consent. These fields are the keys used to build department and university scores, compare states and countries, and follow the same course from one annual wave to the next.

The research team completes this part with the course director (by phone, video call or email). Each form describes one course, placed in its department and university. Saving this part generates the XXX-XXX access code given to the course director.

A01. Country. Country *[required]*

Captures: Country of the school. US schools see the matching US references in later items. Schools in other countries answer the same items, so their scores can be compared with US schools.

- □ United States
- □ Other country

Other, specify: _____

A02. State / region. State (United States) or province / region (other countries) *[required]*

Captures: State or region of the school, to compare schools within a country.

Answer: _____

A03. Institution. Name of dental school (institution) *[required]*

Captures: The university (dental school). The courses of all its departments are combined into the school's benchmark score.

Answer: _____

A04. Institution type. Type of institution *[required]*

Captures: Funding model of the school, to test whether public and private schools differ in maturity and resources.

- □ Public (federal)
- □ Public (state / municipal)
- □ Private, non-profit
- □ Private, for-profit

A05. City. City *[required]*

Captures: City of the campus, for mapping and to distinguish campuses of the same institution.

Answer: _____

A06. Department. Name of the department that teaches the course (e.g., Department of Oral Radiology) *[required]*

Captures: The department. Course scores are first combined at this level.

Answer: _____

A07. Course. Course code and title (e.g., "RAD-301 Oral Radiology II") *[required]*

Captures: The named course, so readers can see exactly which courses produced each school's curriculum results.

Answer: _____

A08. Training level. Training level of this course *[required]*

Captures: Training level, to compare predoctoral, graduate and residency teaching.

- ☐ Predoctoral first professional degree (DDS/DMD)
- ☐ Research master's (MS/MSc)
- ☐ Professional/clinical master's or specialty certificate
- ☐ Doctoral (PhD/DSc)
- ☐ Residency / advanced education (e.g., AEGD, GPR, specialty residency)
- ☐ Integrated long-track program (e.g., DDS/DMD-PhD)
- ☐ Continuing education / CPD
- ☐ Other

Other, specify: _____

A09. Respondent role. Role of the respondent *[required]*

Captures: Who answered. The expected respondent is the course director (the head of the course); another role is recorded when the director delegates. Used to judge how close the respondent is to the information.

- ☐ Department head / chair
- ☐ Course director / coordinator
- ☐ Program director (graduate / residency)
- ☐ Curriculum committee member
- ☐ Faculty representative
- ☐ Dean / associate dean
- ☐ Chief information or informatics officer
- ☐ Other

Other, specify: _____

A10. Respondent name. Respondent full name *[personal data — stored separately]*

Captures: Contact only. Stored in a separate private file, never in the analysis data.

Answer: _____

A11. Respondent email. Respondent email *[personal data — stored separately]*

Captures: Contact only, for reminders and next year's invitation. Stored separately.

Answer: _____

A12. Interviewer. Research team member who completed Part A *[personal data — stored separately]*

Captures: Audit trail of who issued the code. Stored separately.

Answer: _____

A13. Language. Preferred language of the respondent

Captures: Language for communication and translated versions of the instrument.

- ☐ English
- ☐ Spanish
- ☐ Other

Other, specify: _____

A14. Consent. Informed consent to participate was obtained *[required]*

Captures: Ethical requirement. Forms without consent are excluded from analysis.

- ☐ Yes
- ☐ No

A15. Survey year (wave). Survey year (annual wave), e.g., 2026 *[required]*

Captures: The annual wave. The survey is repeated every year so changes in each department can be tracked over time.

Answer (undefined): _____

A16. Previous code. Access code this same course used in the previous year, if any (XXX-XXX)

Captures: Links this year's answers to last year's for the same course, to measure year-to-year progress.

Answer: _____

Part B. Context of the course

Goal. Measure the size of the course, so every count becomes a fair ratio (per student, per faculty member, per year).

What this part captures. Describes the size and structure of the course (students, faculty, calendar, hours) and which areas of health informatics it touches. These numbers are denominators: they turn raw counts (e.g., computers, trained faculty, teaching hours) into comparable ratios.

Answer every item for the course named in Part A. Where an item asks about resources or rules, describe what this course can use or must follow, even if the department or school provides it. Approximate numbers are acceptable.

B01. Enrollment. Approximate number of students enrolled in this course per year

Captures: Size of the student body, used to compute students per computer, per device and per trained teacher.

Answer (students): _____

B02. Faculty headcount. Number of faculty members assigned to this course (headcount)

Captures: Teaching staff size, denominator for the share of faculty with AI, informatics, robotics or bioinformatics expertise.

Answer (faculty): _____

B03. Faculty FTE. Full-time equivalent (FTE) of faculty assigned to this course

Captures: Teaching capacity adjusted for part-time staff.

Answer (FTE): _____

B04. Academic calendar. Academic calendar used by this course

Captures: Calendar type, needed because hours are asked per term and calendars differ between schools.

US reference: see table "US context", row "Academic calendar".

- ☐ Semester (2 terms per year)
- ☐ Trimester (3 terms per year)
- ☐ Quarter (4 terms per year)
- ☐ Annual system (1 term per year)
- ☐ Modular / block system

B05. Terms per year. Number of academic terms per year (used to convert hours per term into hours per year)

Captures: Converts every "hours per term" answer into hours per year so schools are comparable.

Answer (terms): _____

B06. Course hours. Total contact hours of this course per term (all topics)

Captures: Total course time, to express technology teaching as a share of all teaching time.

Answer (hours per term): _____

B07. Program years. In which program year(s) is this course taught?

Captures: When in the program students meet this content (early exposure vs. clinical years).

- ☐ Year 1
- ☐ Year 2
- ☐ Year 3
- ☐ Year 4
- ☐ Year 5
- ☐ Year 6 or later / internship
- ☐ Graduate / residency year

B08. Course format. Main format of this course

Captures: Teaching setting, because equipment and tool use differ between lecture, lab and clinic courses.

- ☐ Didactic (lectures/seminars)
- ☐ Preclinical / laboratory
- ☐ Clinical
- ☐ Mixed
- ☐ Research

B09. Areas of health informatics addressed. Which areas of health informatics does this course address in teaching, research or care?

Captures: Which areas of health informatics are present at all, giving a quick profile before the detailed parts.

- ☐ Health information systems (EHR, imaging systems, teledentistry, data)
- ☐ Artificial intelligence
- ☐ Robotics and digital dentistry (CAD/CAM, simulators, robots, navigation)
- ☐ Bioinformatics and omics
- ☐ Software development for health
- ☐ None of these

B10. Respondent experience. Your own experience with digital technologies (AI, informatics, robotics or bioinformatics) in teaching, research or practice

Captures: Respondent background, used to check whether answers vary with the respondent's own experience.

- ☐ None
- ☐ Less than 1 year
- ☐ 1–3 years
- ☐ More than 3 years

B11. Respondent AI literacy. How would you rate your own AI literacy?

Captures: Self-rated AI literacy of the respondent (not scored as course maturity).

- ☐ (1) Basic: I know general concepts only
- ☐ (2) Intermediate: I can interpret AI outputs and their limitations
- ☐ (3) Advanced: I can critically appraise AI studies and select tools
- ☐ (4) Expert: I develop, train or validate AI models

Part C. Physical resources: computing, AI tools and data

Goal. Know what the course has today: computing, software, AI tools, data, privacy and security.

What this part captures. Pictures the physical and digital resources that every technology domain depends on: computers and GPUs, cloud access, dedicated spaces, software, AI tools actually used by students,

datasets, data quality, privacy, connectivity, interoperability and security. It answers "what does this course have today?"

Facilities, equipment, software, data and technical support available to the named course.

C01. Computing hardware. Is there hardware designated for AI or data-intensive teaching or research (GPU workstations, high-performance computing, cloud credits) in this course? *[domain INF]*

Captures: Whether the course has its own computing power for AI and data work, or depends on personal equipment.

- ☐ (1) None
- ☐ (2) Only individual faculty members' personal equipment
- ☐ (3) Shared equipment, insufficient for demand
- ☐ (4) Dedicated high-performance computing, regularly updated and accessible to students
- ☐ (-1) Don't know / Not applicable

C02. GPU workstations. Number of GPU-equipped workstations or servers available to this course

Captures: Hard count of AI-capable machines, comparable across schools and over years.

Answer (units): _____

C03. Cloud computing. Access to cloud computing (institutional cloud accounts, research credits, university HPC cluster) *[domain INF]*

Captures: Access to scalable computing beyond local machines.

- ☐ (1) No access
- ☐ (2) Individual/ad hoc access paid by faculty
- ☐ (3) Institutional access on request, limited quota
- ☐ (4) Routine institutional access with support
- ☐ (-1) Don't know / Not applicable

C04. Student devices ratio. Number of students per institutionally provided computer during hands-on digital sessions (e.g., 1 = one computer per student)

Captures: Whether students can actually practise hands-on, or only watch.

Answer (students per computer): _____

C05. Dedicated digital space. Is there a dedicated physical space for digital dentistry, informatics or AI teaching/research (lab, innovation hub)? *[domain INF]*

Captures: Existence of a physical home for digital work, a strong sign of institutional commitment.

- ☐ (1) None
- ☐ (2) Shared general computer room
- ☐ (3) Dedicated digital dentistry lab without AI focus
- ☐ (4) Dedicated AI/digital dentistry lab with staff and scheduled student access
- ☐ (-1) Don't know / Not applicable

C06. Software licences. Which data-science and AI software is available to students and faculty of this course?

Captures: The software toolbox available, from programming languages to annotation and commercial AI.

- ☐ Python / Jupyter / Google Colab
- ☐ R / RStudio
- ☐ MATLAB
- ☐ No-code machine learning (e.g., Orange, KNIME, Teachable Machine)
- ☐ Image annotation tools (e.g., CVAT, Label Studio, 3D Slicer)
- ☐ Statistical packages (SPSS, Stata, SAS)

- ☐ Commercial dental AI platforms
- ☐ Institutionally licensed generative AI (e.g., Copilot, ChatGPT Edu, Gemini)
- ☐ None

Other, specify: _____

C07. AI-enabled equipment. Which AI-enabled equipment or systems are in use in this course?

Captures: Where AI is already embedded in equipment, often without being taught explicitly.

- ☐ Digital radiography with AI analysis
- ☐ CBCT with AI segmentation
- ☐ Intraoral scanners with AI features
- ☐ CAD/CAM or 3D printing with AI design
- ☐ Haptic / simulation units with AI feedback
- ☐ Electronic health record with AI modules
- ☐ Learning management system with AI features
- ☐ AI tutor / chatbot for students
- ☐ None

Other, specify: _____

C08. AI tools used by students. List each AI tool students actually use in this course. Leave rows empty if not applicable.

Captures: An inventory of real AI tools: what they do, whether they are approved, how deeply students use them and since when. Pooled across schools it becomes a map of dental AI adoption.

#	Tool / product name	Main task	Regulatory status	Student use	Part of graded work?	Year introduced
1						
2						
3						
4						
5						
6						

Main task codes: caries = Caries detection on radiographs; perio = Periodontal bone-loss assessment; periapical = Periapical lesion detection; cbct = CBCT segmentation / 3D analysis; ceph = Cephalometric landmarking / orthodontic analysis; implant = Implant planning / guided surgery; cad = CAD/CAM restoration design; ios = Intraoral-scan analysis / smile design; lesion = Oral mucosal lesion / oral cancer screening; charting = Charting, clinical notes, ambient scribe; patient = Patient communication / education; admin = Scheduling, billing, administration; sim = Preclinical simulation feedback / preparation grading; llm = General-purpose LLM / chatbot; other = Other

Regulatory status codes: 1 = Cleared/approved by the national regulator for this use; 2 = Foreign approval only (e.g., CE, FDA); 3 = Commercial, not cleared; 4 = Research / in-house prototype; -1 = Don't know

Student use codes: 1 = Students observe faculty use; 2 = Preclinical / simulation only; 3 = Students operate under supervision on patients; 4 = Students operate independently in patient care

Part of graded work? codes: 1 = Yes; 0 = No

C09. Number of AI tools. Overall use of AI tools by students in this course *[domain INF]*

Captures: Summary of AI tool adoption for scoring.

- ☐ (1) No AI tool is used
- ☐ (2) 1 AI tool used
- ☐ (3) 2–3 AI tools used

- ☐ (4) 4 or more AI tools used and integrated into graded coursework
- ☐ (-1) Don't know / Not applicable

C10. De-identified dataset. Is there a standardized, de-identified oral-health dataset (radiographs, intraoral scans, treatment records) that students or faculty can access for AI training or research? *[domain DAT]*

Captures: Availability of data to train and test AI, the key ingredient for the consortium's open AI model.

- ☐ (1) No dataset accessible
- ☐ (2) Dataset in planning or pilot stage
- ☐ (3) Dataset exists but access is restricted and scale is limited
- ☐ (4) Dataset exists with clear access protocols and is used in the course
- ☐ (-1) Don't know / Not applicable

C11. Dataset modalities. Which data modalities are available in that dataset?

Captures: Which data types exist, to plan which parts of a shared AI model each school could support.

- ☐ Periapical / bitewing radiographs
- ☐ Panoramic radiographs
- ☐ Cephalometric radiographs
- ☐ CBCT
- ☐ Intraoral scans (3D)
- ☐ Clinical photographs
- ☐ Histopathology images
- ☐ Structured treatment / EHR records
- ☐ Free-text clinical notes
- ☐ No dataset

C12. Dataset size. Approximate number of de-identified images or records in that dataset (enter 0 if none)

Captures: Scale of existing data.

Answer (images/records): _____

C13. Annotation capacity. Capacity to label/annotate data for AI (calibrated annotators, annotation protocol, inter-rater agreement) *[domain DAT]*

Captures: Ability to produce reliable labels, which determines whether data can be used to train trustworthy AI.

- ☐ (1) No capacity
- ☐ (2) Ad hoc annotation by individual researchers
- ☐ (3) Annotation protocol exists, limited calibration
- ☐ (4) Calibrated annotators with documented protocol and agreement statistics
- ☐ (-1) Don't know / Not applicable

C14. Data quality. What measures ensure the accuracy, completeness and timeliness of data used in this course? *[domain DAT]*

Captures: Data quality assurance (PAHO/WHO readiness toolkit item).

- ☐ (1) No measures applied
- ☐ (2) Ad hoc checks only
- ☐ (3) Some measures exist but are not comprehensive
- ☐ (4) Systematic quality assurance with documented protocols
- ☐ (-1) Don't know / Not applicable

C15. Data representativeness. Is the representativeness of the data assessed (age, sex, ethnicity, socioeconomic group, rural/urban, device type)? *[domain DAT]*

Captures: Whether datasets reflect the patients the AI will serve (PAHO bias-free AI guidance).

- ☐ (1) Never assessed
- ☐ (2) Discussed informally
- ☐ (3) Assessed for some projects
- ☐ (4) Systematically assessed and reported
- ☐ (-1) Don't know / Not applicable

C16. Privacy and security. What privacy and security systems protect patient data used in this course? *[domain DAT]*

Captures: Protection of patient data, read against US privacy law (HIPAA, FERPA).

US reference: see table "US context", row "Privacy law".

- ☐ (1) No systems
- ☐ (2) Basic password protection only
- ☐ (3) Systems exist but are inconsistent
- ☐ (4) Comprehensive systems with encryption, audit trails and a named person responsible
- ☐ (-1) Don't know / Not applicable

C17. De-identification. Are there documented procedures for anonymization / de-identification of patient data before use in teaching or research? *[domain DAT]*

Captures: Readiness to share data safely, a precondition for joint AI development.

- ☐ (1) No procedures
- ☐ (2) Informal practice only
- ☐ (3) Written procedures, inconsistently applied
- ☐ (4) Written procedures, routinely applied and audited
- ☐ (-1) Don't know / Not applicable

C18. Secure research environment. Is a secure environment available for working with identifiable or sensitive data (trusted research environment, approved server, data enclave)? *[domain DAT]*

Captures: Whether sensitive data can be analysed without leaving a protected environment.

- ☐ (1) Not available
- ☐ (2) Planned
- ☐ (3) Available on request with delays
- ☐ (4) Available and routinely used
- ☐ (-1) Don't know / Not applicable

C19. Connectivity and IT support. Does this course have reliable broadband and technical support for the digital tools it uses? *[domain INF]*

Captures: Basic connectivity and support, without which equipment goes unused.

- ☐ (1) Neither reliable connectivity nor specific support
- ☐ (2) Basic IT support exists but does not cover these tools
- ☐ (3) Partial support (software installation only)
- ☐ (4) Dedicated educational-technology support assigned to the unit
- ☐ (-1) Don't know / Not applicable

C20. Interoperability. Can the digital tools used exchange data with existing systems (EHR, imaging/PACS, learning management system) using standards such as DICOM, HL7 FHIR or LTI? *[domain INF]*

Captures: Whether systems talk to each other, a core health-informatics capability.

- ☐ (1) No interoperability
- ☐ (2) Some standards under development
- ☐ (3) Standards exist but implementation is inconsistent
- ☐ (4) Well-established standards enabling seamless integration
- ☐ (-1) Don't know / Not applicable

C21. Cybersecurity. Are cybersecurity measures applied to digital tools and data (risk assessment, access control, vendor security review)? *[domain INF]*

Captures: Security of systems that hold patient data.

- ☐ (1) None
- ☐ (2) Basic institutional measures only
- ☐ (3) Measures exist but not specific to these tools
- ☐ (4) Specific security review before adoption and periodic audit
- ☐ (-1) Don't know / Not applicable

C22. Equitable student access. Do all students in this course have equal access to the tools and computing needed (including outside class hours)? *[domain INF]*

Captures: Equity of access among students.

- ☐ (1) No — access depends on personal devices/subscriptions
- ☐ (2) Limited institutional access
- ☐ (3) Access during scheduled sessions only
- ☐ (4) Equal access during and outside class hours
- ☐ (-1) Don't know / Not applicable

Part D. Human resources and expertise (AI and data science)

Goal. Know who can teach, build and support these technologies, and what holds them back.

What this part captures. Pictures the people: how many faculty can teach AI, who does AI research, what training they received, which partners and technical staff support them, and what holds them back. Equipment without trained people does not change teaching; this part measures that capacity.

Faculty and staff assigned to the named course.

D01. Faculty able to teach AI. What proportion of faculty can independently teach AI content (machine-learning principles, interpreting AI outputs, critical appraisal of AI literature)? *[domain HUM]*

Captures: Teaching capacity for AI as a share of all faculty.

- ☐ (1) 0%
- ☐ (2) 1–10%
- ☐ (3) 11–25%
- ☐ (4) More than 25%
- ☐ (-1) Don't know / Not applicable

D02. Faculty teaching AI (n). Number of faculty who currently teach any AI content in this course

Captures: Exact count, comparable across years.

Answer (faculty): _____

D03. Faculty with formal AI degree. Number of faculty with a formal qualification in AI, computer science, data science or biomedical informatics (degree or university certificate)

Captures: Deep expertise in the course and its department, beyond self-taught knowledge.

Answer (faculty): _____

D04. AI-primary researchers. How many faculty have AI in dentistry or dental education as their primary research area? *[domain HUM]*

Captures: Research leadership in AI, often the engine of curricular change.

- ☐ (1) None
- ☐ (2) 1
- ☐ (3) 2–3
- ☐ (4) 4 or more
- ☐ (-1) Don't know / Not applicable

D05. AI publications (3 y). Number of peer-reviewed AI-related publications by faculty of this course in the past 3 years

Captures: Research output, a verifiable indicator of AI activity.

Answer (publications): _____

D06. AI graduate trainees. Number of graduate students, residents or postdocs currently working on AI projects in this course

Captures: The next generation of AI researchers being trained.

Answer (trainees): _____

D07. Faculty AI training. What formal AI training have faculty received in the past 24 months? *[domain HUM]*

Captures: Institutional investment in faculty development.

- ☐ (1) No training received
- ☐ (2) Voluntary, sporadic sessions only
- ☐ (3) Organized training, but low or inconsistent attendance
- ☐ (4) Systematic training with institutional incentives and tracking
- ☐ (-1) Don't know / Not applicable

D08. Training hours per faculty. Average hours of AI training per faculty member in the past 12 months

Captures: Intensity of faculty training.

Answer (hours): _____

D09. Training formats. Formats of AI training available to faculty

Captures: How faculty learn, to design consortium training that fills gaps.

- ☐ Institutional workshops / seminars
- ☐ Online courses (MOOCs, e.g., Coursera, edX)
- ☐ University certificate programs
- ☐ Conference sessions (IADR, ADEA, ADEE, SBPqO, IDA, CSA)
- ☐ Peer mentoring / communities of practice
- ☐ Vendor training
- ☐ Sponsored degree programs
- ☐ None

Other, specify: _____

D10. Interdisciplinary partners. Name each external unit this course collaborates with on AI, informatics, robotics or bioinformatics (e.g., Department of Computer Science, Faculty of Engineering, a company).

Captures: The collaboration network, showing who dental schools rely on for technical expertise.

#	Partner unit	Type
1		
2		
3		

Type codes: 1 = Same university — computer science / engineering; 2 = Same university — medicine / public health / biology / other; 3 = Other university (national); 4 = Other university (international); 5 = Industry / company; 6 = Government / hospital / other

D11. Partnership depth. Depth of interdisciplinary collaboration *[domain HUM]*

Captures: How formal and durable those partnerships are.

- ☐ (1) No partners
- ☐ (2) Informal individual collaboration only
- ☐ (3) Formal project-based collaboration
- ☐ (4) Long-term institutional partnership with joint funding or appointments
- ☐ (-1) Don't know / Not applicable

D12. Technical staff. Access to dedicated technical personnel (data scientists, AI engineers, statisticians, research software engineers) *[domain HUM]*

Captures: Technical support available to faculty and students.

- ☐ (1) No access
- ☐ (2) Shared across the school, high demand
- ☐ (3) Adequate access for most needs
- ☐ (4) Dedicated technical staff embedded in the department
- ☐ (-1) Don't know / Not applicable

D13. Technical staff FTE. Full-time equivalent of technical data/AI staff available to this course

Captures: Exact technical staffing, comparable over years.

Answer (FTE): _____

D14. Hiring plans. Is recruitment of faculty or staff with AI, informatics, robotics or bioinformatics expertise planned? *[domain HUM]*

Captures: Forward-looking investment in people.

- ☐ (1) Not planned
- ☐ (2) Discussed, no approved position
- ☐ (3) Position approved or recruitment underway
- ☐ (4) Expertise already recruited and further positions planned
- ☐ (-1) Don't know / Not applicable

D15. Faculty openness. Overall attitude of faculty in this course toward integrating these technologies *[domain HUM]*

Captures: Cultural readiness for change.

- ☐ (1) Mostly resistant
- ☐ (2) Mixed / indifferent
- ☐ (3) Mostly supportive

- ☐ (4) Strongly supportive with active champions
- ☐ (-1) Don't know / Not applicable

D16. Faculty competency barriers. Most significant barriers to faculty competency (select up to two)

Captures: The main obstacles, so consortium support can target them.

Select up to 2.

- ☐ Lack of faculty time
- ☐ Lack of funding for training
- ☐ Lack of faculty interest
- ☐ Lack of qualified trainers
- ☐ Competing curricular priorities
- ☐ No recognition in promotion/tenure
- ☐ Training not available in local language

Other, specify: _____

Part E. AI curriculum and teaching

Goal. Know what students actually learn about AI, at which level and for how many hours.

What this part captures. Measures what students are actually taught about AI and for how long: hours per term by format and by 16 topics (including programming and formal model development), named undergraduate and graduate courses, assessment, materials and planned changes. Hours are converted to hours per year (with B05) to build an international benchmark.

Refers to the course named in Part A. Hours: include lectures, hands-on sessions and assignments; exclude generic literature searching and routine use of clinical software.

E01. Dedicated AI instruction. Does the course include dedicated AI instruction (not a passing mention)? *[domain CUR]*

Captures: Whether AI is taught deliberately, not only mentioned.

- ☐ (1) None
- ☐ (2) Mentioned in 1–2 lectures only
- ☐ (3) 1 dedicated module or class session (≤4 hours)
- ☐ (4) 1 or more dedicated AI courses, or systematic integration across sessions
- ☐ (-1) Don't know / Not applicable

E02. AI hours by format. Hours of AI-specific instruction a student receives in this course per term, by format (enter 0 if none)

Captures: Total AI teaching time and how much of it is hands-on rather than passive.

Topic / format	Hours (hours per term)
Lectures / seminars	
Hands-on / computer lab	
Clinical / preclinical use with supervision	
Assignments / guided self-study	

E03. AI hours by topic. Hours per term devoted to each AI topic in this course (enter 0 if not covered; a session may count toward more than one topic)

Captures: The content profile of AI teaching across 16 topics, from fundamentals to programming, validation, bias and regulation.

Topic / format	Hours (hours per term)
AI fundamentals and terminology	
Machine learning / deep learning principles	
Data science and statistics for AI	
Programming (Python, R)	
Model development and training	
Model validation and performance metrics (sensitivity, AUC, calibration)	
Bias, fairness and subgroup performance	
Explainability / interpretability	
Clinical interpretation of AI outputs and AI errors	
AI in dental imaging (radiographs, CBCT)	
AI in treatment planning, CAD/CAM, orthodontics, implants	
Generative AI, large language models and prompt design	
Ethics, privacy, informed consent, accountability	
Regulation, standards and approval of AI devices	
Critical appraisal of AI literature and reporting guidelines	
AI as a research tool (literature, analysis, writing)	

E04. Required or elective. Is the AI content of this course required or elective?

Captures: Whether every graduate is exposed, or only volunteers.

- ☐ No AI content
- ☐ Elective / optional
- ☐ Required for some students / tracks
- ☐ Required for all students

E05. AI credits. Academic credits awarded for AI content (local credit units; 0 if none)

Captures: Formal weight of AI in the curriculum.

Answer (credits): _____

E06. Technical depth. Which best describes the technical depth of AI instruction? *[domain CUR]*

Captures: Depth of AI teaching, from awareness to building models.

- ☐ (1) Conceptual awareness only
- ☐ (2) Clinical interpretation of AI outputs
- ☐ (3) Hands-on operation of existing AI tools under guidance
- ☐ (4) Includes programming or formal AI development
- ☐ (-1) Don't know / Not applicable

E07. Programming / AI development. Does the course cover programming (Python/R) or formal AI development (model training, validation, bias assessment)? *[domain CUR]*

Captures: Whether students gain the skills needed to contribute to building AI.

- ☐ (1) None
- ☐ (2) Instructor demonstration only
- ☐ (3) Hands-on exercises under guidance
- ☐ (4) Students independently complete a small AI project (e.g., train and validate a classifier)
- ☐ (-1) Don't know / Not applicable

E08. Development platforms. Platforms used for programming or model development

Captures: The tools used for hands-on development.

- ☐ Python / Jupyter / Colab
- ☐ R
- ☐ No-code tool (Orange, KNIME, Teachable Machine)
- ☐ Cloud AutoML (Google, Azure, AWS)
- ☐ Vendor research platform
- ☐ Not applicable

Other, specify: _____

E09. Programming hours. Hours per term devoted to programming or model development

Captures: Time spent on programming and formal AI development specifically.

Answer (hours per term): _____

E10. Student AI projects. Number of student AI projects (coursework, theses, capstones) completed in the past academic year

Captures: Concrete student output in AI.

Answer (projects): _____

E11. Undergraduate AI courses. Name the undergraduate course(s) in this course that are dedicated to AI or contain a dedicated AI module

Captures: The named undergraduate AI courses and their hours, for the transparency list in publications.

#	Course code and title	AI hours per term	Status
1			
2			
3			

Status codes: 1 = Elective; 2 = Required

E12. Graduate / residency AI courses. Name the graduate or residency course(s) in this course that include AI content

Captures: The named graduate and residency AI courses and their hours.

#	Course code and title	AI hours per term	Status
1			
2			
3			

Status codes: 1 = Elective; 2 = Required

E13. Graduate AI offering. Graduate-level AI offering in this course *[domain CUR]*

Captures: Maturity of AI at graduate level.

- ☐ (1) No graduate AI course
- ☐ (2) Elective only
- ☐ (3) Required course exists
- ☐ (4) Required course plus a formal AI research project requirement
- ☐ (-1) Don't know / Not applicable

E14. AI theses (3 y). Number of master's/doctoral theses or residency projects on AI defended in this course in the past 3 years

Captures: Graduate research output in AI.

Answer (theses): _____

E15. Clinical AI training. Are students trained in clinical application and critical interpretation of AI outputs (reading AI-assisted reports, identifying AI errors, understanding limitations)? *[domain CUR]*

Captures: Preparation for safe use of AI with patients.

- ☐ (1) None
- ☐ (2) Informal discussion only
- ☐ (3) 1 dedicated session
- ☐ (4) Integrated into clinical rotations with formal assessment
- ☐ (-1) Don't know / Not applicable

E16. Assessment of AI competency. How is student AI competency assessed? *[domain CUR]*

Captures: Whether AI learning is verified, not just delivered.

- ☐ (1) Not assessed
- ☐ (2) Written exam only
- ☐ (3) Separate AI assignment or project
- ☐ (4) Competency-based assessment (OSCE station, portfolio, practical demonstration)
- ☐ (-1) Don't know / Not applicable

E17. Teaching methods. Teaching methods used for AI content

Captures: How AI is taught, to identify practices worth sharing across the consortium.

- ☐ Lectures
- ☐ Flipped classroom
- ☐ Case-based learning
- ☐ Problem-based learning
- ☐ Journal club / critical appraisal
- ☐ Computer lab / coding exercises
- ☐ Hackathon / competition
- ☐ Supervised clinical use of AI tools
- ☐ Simulation with AI feedback
- ☐ AI tutor / chatbot-assisted learning

Other, specify: _____

E18. Educational materials. Dedicated AI educational materials (textbooks, online modules, case libraries) *[domain CUR]*

Captures: Quality of teaching resources.

- ☐ (1) None used

- ☐ (2) Scattered resources, not formally organized
- ☐ (3) Some dedicated materials used
- ☐ (4) Comprehensive, curated collection integrated into the course
- ☐ (-1) Don't know / Not applicable

E19. Materials used. Name the main AI educational materials used

Captures: Specific materials, to build a shared open library.

#	Material (title / URL)
1	
2	
3	

E20. Competency framework. Is AI teaching aligned with a competency framework (e.g., IADR/ITU-WHO core curriculum for dental AI, ADEA competencies, CODA standards)? *[domain CUR]*

Captures: Alignment with curriculum guidance and accreditation standards.

US reference: see table "US context", row "Curriculum / accreditation body".

- ☐ (1) No framework
- ☐ (2) Aware of frameworks, not used
- ☐ (3) Partially aligned
- ☐ (4) Explicitly mapped to a framework with defined learning outcomes
- ☐ (-1) Don't know / Not applicable

E21. AI as a teaching tool. Is AI used as a tool to deliver teaching (adaptive learning, automated feedback on preparations, AI-generated cases, virtual patients)? *[domain CUR]*

Captures: AI used to teach, not only taught as a topic.

- ☐ (1) Not used
- ☐ (2) Experimented by individual faculty
- ☐ (3) Used regularly in some activities
- ☐ (4) Systematically used and evaluated
- ☐ (-1) Don't know / Not applicable

E22. CPD AI offerings. Number of continuing-education (CPD) activities on AI offered by this course in the past 12 months

Captures: Reach of the department beyond its own students, to practising dentists.

Answer (activities): _____

E23. Student preparedness. How prepared and interested in AI are students in this course? *[domain CUR]*

Captures: Student readiness as perceived by faculty.

- ☐ (1) Low interest and preparedness
- ☐ (2) Interest but low preparedness
- ☐ (3) Interest and moderate preparedness
- ☐ (4) High interest and preparedness
- ☐ (-1) Don't know / Not applicable

E24. Planned curriculum change. Is an increase of AI content planned for the next 2 years? *[domain CUR]*

Captures: Momentum: where the curriculum is heading.

- ☐ (1) No plans

- ☐ (2) Under informal discussion
- ☐ (3) Approved, not yet implemented
- ☐ (4) Being implemented now
- ☐ (-1) Don't know / Not applicable

Part F. Generative AI in teaching and learning

Goal. Know whether generative AI is used with clear rules, real skills and working safeguards.

What this part captures. Captures how the course governs and teaches generative AI: policies, permitted uses, prompt-design and verification skills, and safeguards (adapted from PAHO "AI prompt design" and AI-GUARD Tier 1). Generative AI spread faster than any earlier technology, so this part is tracked closely year to year.

Generative AI includes chatbots and large language models such as ChatGPT, Gemini, Copilot, DeepSeek, Qwen or Ernie Bot.

F01. Generative AI policy. Which policy governs student and faculty use of generative AI in this course? *[domain GEN]*

Captures: Maturity of generative-AI rules.

- ☐ (1) No policy
- ☐ (2) General academic-integrity policy only, not AI-specific
- ☐ (3) Interim guidance issued
- ☐ (4) Formal, regularly updated policy with training
- ☐ (-1) Don't know / Not applicable

F02. Policy name. Name of the policy document (and level: university / school / department / course)

Captures: The actual document, so policies can be compared and shared.

Answer: _____

F03. Permitted student uses. Which student uses of generative AI are permitted in this course?

Captures: What students may actually do with generative AI.

- ☐ Studying / explanations
- ☐ Writing support with disclosure
- ☐ Practising clinical cases
- ☐ Coding assistance
- ☐ Literature search / research support
- ☐ Use during assessments
- ☐ All use is prohibited
- ☐ Not defined

Other, specify: _____

F04. Prompt design teaching. Are students taught how to design effective prompts (context, task, format, audience, constraints)? *[domain GEN]*

Captures: Prompt design as a core digital skill (PAHO).

- ☐ (1) Not taught
- ☐ (2) Mentioned informally
- ☐ (3) 1 dedicated session
- ☐ (4) Integrated practice with feedback
- ☐ (-1) Don't know / Not applicable

F05. Verification of outputs. Are students taught to verify generative-AI outputs (hallucinations, fabricated references, outdated guidance)? *[domain GEN]*

Captures: Critical checking of AI output, essential for patient safety.

- ☐ (1) Not taught
- ☐ (2) Mentioned informally
- ☐ (3) 1 dedicated session
- ☐ (4) Integrated practice with assessment
- ☐ (-1) Don't know / Not applicable

F06. Generative AI safeguards. Which safeguards are in place for generative AI use in this course? (adapted from PAHO AI-GUARD, Tier 1)

Captures: A safeguards checklist scored Yes = 2, Partial = 1, No = 0 into a generative-AI safeguards index.

Item	Yes	Partial	No	Don't know
Rule that no identifiable patient data is entered into public AI tools	☐	☐	☐	☐
An institutionally licensed, privacy-protected generative AI tool is available	☐	☐	☐	☐
Students must disclose AI use in assignments	☐	☐	☐	☐
Human verification required before AI output is used in patient care	☐	☐	☐	☐
Process to report misuse or AI errors	☐	☐	☐	☐
Staff and students trained on limitations and bias of generative AI	☐	☐	☐	☐

Scoring: Yes = 2, Partial = 1, No = 0.

F07. Faculty GenAI use. How often do faculty use generative AI to prepare teaching materials or assessments?

Captures: Faculty adoption of generative AI in their own work.

- ☐ Never
- ☐ Occasionally
- ☐ Monthly
- ☐ Weekly or more
- ☐ Don't know

F08. AI-detection software. Is AI-text detection software used to assess student work?

Captures: Reliance on detection tools, which have known accuracy problems.

- ☐ Yes, routinely
- ☐ Yes, occasionally
- ☐ No
- ☐ Don't know

Part G. Health information systems and software development

Goal. Know whether the clinic produces reliable digital data, and whether the course can build and maintain its own health software.

What this part captures. Measures the digital backbone of the clinic and how informatics is taught: electronic records, imaging archives, coding, teledentistry, analytics and research data access, plus teaching hours and staff. It also measures software development for health: whether the course builds, tests and maintains its own tools, and whether students learn to do so. Information systems produce the structured data that AI and research depend on.

Health informatics is the umbrella of this survey: AI, robotics, bioinformatics and software development for health all sit inside it. This part covers its core, health information systems (electronic health records, imaging systems, coding and terminologies, teledentistry, data analytics, geographic information systems), and software development for health (G13–G16).

G01. Electronic health record. Which best describes the patient record system used in the clinics where this course is taught? *[domain HIN]*

Captures: Digital maturity of patient records, the main source of clinical data.

- ☐ (1) Paper records
- ☐ (2) Electronic scheduling/billing only; clinical notes on paper
- ☐ (3) Electronic clinical records, mostly free text
- ☐ (4) Integrated electronic record with structured data, imaging and decision support
- ☐ (-1) Don't know / Not applicable

G02. EHR system name. Name the electronic record and imaging (PACS) systems used (e.g., axiUm, Dentrix, MiPACS)

Captures: The actual systems, to study interoperability and shared data formats across schools.

Answer: _____

G03. Coding and terminologies. Which coding systems are used to record diagnoses and procedures?

Captures: Use of standard codes, which makes data comparable and usable for AI.

- ☐ ICD-10 / ICD-11
- ☐ SNODENT / SNOMED CT
- ☐ Procedure codes (CDT, TUSS or national equivalent)
- ☐ Free text only / none

Other, specify: _____

G04. Digital imaging archive. How are dental images stored and shared? *[domain HIN]*

Captures: Whether images are centrally available, a precondition for imaging AI.

- ☐ (1) Film or images not archived centrally
- ☐ (2) Digital images stored per device or computer
- ☐ (3) Central archive (PACS) for some modalities
- ☐ (4) Enterprise PACS integrated with the health record via DICOM
- ☐ (-1) Don't know / Not applicable

G05. Teledentistry. Use of teledentistry (remote consultation, triage, monitoring) in this course *[domain HIN]*

Captures: Remote care capacity and student exposure to it.

- ☐ (1) None
- ☐ (2) Ad hoc use by individuals
- ☐ (3) Structured program in some services
- ☐ (4) Routine teledentistry with student involvement
- ☐ (-1) Don't know / Not applicable

G06. Data analytics. Are clinic and teaching data analysed with dashboards or routine reports? *[domain HIN]*

Captures: Use of data for decisions, a sign of informatics maturity.

- ☐ (1) Never
- ☐ (2) Occasional manual reports
- ☐ (3) Regular reports for some indicators
- ☐ (4) Routine dashboards used for clinical and teaching quality decisions

- ☐ (-1) Don't know / Not applicable

G07. Research access to clinical data. Can researchers access clinic data (de-identified) for research? *[domain HIN]*

Captures: Ability to turn routine care into research and AI datasets.

- ☐ (1) Not possible
- ☐ (2) Ad hoc extracts by IT
- ☐ (3) Formal request process
- ☐ (4) Curated research data warehouse with governance
- ☐ (-1) Don't know / Not applicable

G08. Public-health informatics / GIS. Use of population health informatics (geographic information systems, surveillance data, epidemiological databases) in teaching or research *[domain HIN]*

Captures: Use of population-level and geographic data in oral public health.

- ☐ (1) Never
- ☐ (2) Occasionally
- ☐ (3) In some projects or courses
- ☐ (4) Routinely
- ☐ (-1) Don't know / Not applicable

G09. Health informatics hours by topic. Hours per term of health-informatics teaching in this course, by topic (enter 0 if none)

Captures: Time spent teaching health informatics, by topic, converted to hours per year.

Topic / format	Hours (hours per term)
Electronic records and documentation quality	
Coding, terminologies and interoperability standards	
Teledentistry and digital communication	
Data analytics, epidemiology and GIS	
Privacy, cybersecurity and data protection law	

G10. Informatics teaching depth. Depth of health-informatics teaching *[domain HIN]*

Captures: Whether informatics is a real part of the curriculum.

- ☐ (1) None
- ☐ (2) Mentioned in lectures
- ☐ (3) Dedicated module
- ☐ (4) Dedicated course with practical assessment
- ☐ (-1) Don't know / Not applicable

G11. Faculty with informatics training. Number of faculty with formal training in health informatics or GIS

Captures: Human capacity in health informatics.

Answer (faculty): _____

G12. Clinical informatics staff. Support from clinical informatics staff (EHR analysts, imaging/PACS administrators) *[domain HIN]*

Captures: Technical staff who keep clinical systems running and data usable.

- ☐ (1) None

- □ (2) Central university IT only
- □ (3) Shared clinical informatics staff
- □ (4) Dedicated clinical informatics team for the school
- □ (-1) Don't know / Not applicable

G13. Software development capacity. Does this course (or its department) develop its own software for health, such as apps, teaching tools, scripts or data pipelines? *[domain SWD]*

Captures: Whether the course can turn ideas into working health software, a core skill of health informatics.

- □ (1) No software developed
- □ (2) Occasional scripts or prototypes by individuals
- □ (3) Some tools built for teaching or research, not maintained
- □ (4) Software built and maintained by a team, with documentation and real users
- □ (-1) Don't know / Not applicable

G14. Development practices. Which practices are used when software is developed? *[domain SWD]*

Captures: Quality and safety of the software that is built.

- □ (1) No software is developed
- □ (2) Ad hoc (no version control or testing)
- □ (3) Version control and some testing
- □ (4) Version control, testing, code review, and security and privacy by design
- □ (-1) Don't know / Not applicable

G15. Teaching software development. Are students taught to design or build health software (requirements, user-centred design, programming, interoperability standards such as HL7 FHIR and DICOM)? *[domain SWD]*

Captures: Whether graduates can take part in building health software.

- □ (1) Not taught
- □ (2) Mentioned only
- □ (3) One dedicated session or module
- □ (4) Structured course or project-based learning with real users
- □ (-1) Don't know / Not applicable

G16. Health software built (3 y). Number of software tools or apps for health developed by this course or its department in the past 3 years

Captures: Output of software development, counted the same way every year.

Answer (tools): _____

Part H. Robotics and digital dentistry

Goal. Know what robotic and digital equipment exists, and whether students really use it.

What this part captures. Takes an inventory of the physical equipment for robotics and digital dentistry (how many units of each), how students access it, how digital workflows are used, who can teach and maintain it, teaching hours and research. Counted every year, it shows how equipment and skills spread across schools.

Includes robot-assisted surgery, haptic and virtual-reality simulators, navigation, scanners, CAD/CAM milling and 3D printing.

H01. Equipment inventory. Number of units available to this course (enter 0 if none)

Captures: A hard count of equipment, comparable across schools and years and convertible to students per unit.

Item	Number (units available)
Robot-assisted implant surgery systems	
Other surgical, endodontic or automated robotic systems	
Dynamic navigation systems for guided surgery	
Haptic or virtual-reality dental simulators	
Conventional simulation units (phantom heads)	
Intraoral scanners	
CBCT units	
Chairside or lab milling units (CAD/CAM)	
3D printers	
Augmented/virtual-reality headsets	

H02. Equipment age. Average age of the main digital and robotic equipment

Captures: How up to date the equipment is.

- ☐ Less than 3 years
- ☐ 3–5 years
- ☐ 6–10 years
- ☐ More than 10 years
- ☐ Don't know

H03. Student access to robotics. Student access to robotic, haptic or navigation systems *[domain ROB]*

Captures: Whether students actually use the equipment.

- ☐ (1) No access
- ☐ (2) Demonstration only
- ☐ (3) Supervised hands-on practice in preclinic
- ☐ (4) Regular hands-on practice including supervised clinical cases
- ☐ (-1) Don't know / Not applicable

H04. Digital workflow. Use of a full digital workflow (scan, design, manufacture) in teaching and clinics *[domain ROB]*

Captures: Integration of digital dentistry into everyday care and teaching.

- ☐ (1) Analog only
- ☐ (2) Partial digital steps
- ☐ (3) Full digital workflow in some clinics
- ☐ (4) Full digital workflow routine across clinics and teaching
- ☐ (-1) Don't know / Not applicable

H05. Robotics/digital hours by topic. Hours per term of robotics and digital-dentistry teaching in this course, by topic (enter 0 if none)

Captures: Time spent teaching robotics and digital dentistry, by topic.

Topic / format	Hours (hours per term)
Digital workflow, CAD/CAM and 3D printing	
Haptic / virtual-reality simulation	
Robot-assisted surgery (principles, demonstration, practice)	
Navigation and guided surgery	
Engineering, automation and maintenance basics	

H06. Faculty trained in robotics. Number of faculty trained to operate or teach robotic, navigation or haptic systems

Captures: Teaching capacity for robotics and digital dentistry.

Answer (faculty): _____

H07. Technical support FTE. Full-time equivalent of technicians or engineers supporting digital and robotic equipment

Captures: Staff who keep the equipment working.

Answer (FTE): _____

H08. Maintenance and replacement. Maintenance and replacement of digital/robotic equipment *[domain ROB]*

Captures: Whether investment in equipment is sustainable.

- ☐ (1) No plan
- ☐ (2) Ad hoc repairs
- ☐ (3) Service contracts for some equipment
- ☐ (4) Full maintenance and replacement plan with budget
- ☐ (-1) Don't know / Not applicable

H09. Safety and credentialing. Training and credentialing protocols before robotic or navigation systems are used on patients *[domain ROB]*

Captures: Patient safety when new technology is used in the clinic.

- ☐ (1) None
- ☐ (2) Informal supervision
- ☐ (3) Written protocol, not always applied
- ☐ (4) Formal credentialing with documented training and supervision
- ☐ (-1) Don't know / Not applicable

H10. Robotics research. Robotics and digital-dentistry research in this course *[domain ROB]*

Captures: Research activity that drives innovation.

- ☐ (1) None
- ☐ (2) Individual projects
- ☐ (3) Funded projects
- ☐ (4) Established research group with engineering partners
- ☐ (-1) Don't know / Not applicable

Part I. Bioinformatics and omics

Goal. Know whether the course can produce, analyse and share biological data.

What this part captures. Pictures laboratory and computing resources for omics research (sequencing, biobanks, pipelines), the people with bioinformatics skills, teaching hours and data-sharing practices. It shows which schools can contribute biological data to future multi-modal AI that links images, records and biology.

Bioinformatics covers the computational analysis of biological data such as the oral microbiome, genomes, gene expression, proteins and metabolites.

I01. Laboratory resources. Which resources are available to this course?

Captures: The physical infrastructure for generating and analysing biological data.

- ☐ Molecular biology laboratory
- ☐ DNA/RNA sequencer in the department or school
- ☐ University sequencing / genomics core facility
- ☐ Biobank of oral samples (saliva, plaque, tissue)
- ☐ Mass spectrometry (proteomics / metabolomics)
- ☐ High-performance computing for bioinformatics
- ☐ None

Other, specify: _____

I02. Biobank and sample governance. Status of biological sample collection and storage *[domain BIO]*

Captures: Ability to run omics studies and link biology to clinical data.

- ☐ (1) None
- ☐ (2) Samples collected per project only
- ☐ (3) Biobank with consent protocol
- ☐ (4) Biobank with linked clinical data, governance and use by several projects
- ☐ (-1) Don't know / Not applicable

I03. Research areas. Bioinformatics and omics research areas active in this course

Captures: Which omics fields the department works in, to match partners across the consortium.

- ☐ Oral microbiome
- ☐ Host genomics / genetic association studies
- ☐ Transcriptomics
- ☐ Proteomics / metabolomics (e.g., saliva)
- ☐ Epigenetics
- ☐ Craniofacial and developmental genetics
- ☐ Antimicrobial resistance
- ☐ None

Other, specify: _____

I04. Software and pipelines. Bioinformatics software and resources used

Captures: Computational tools in use, and whether analyses are reproducible.

- ☐ QIIME 2 / mothur (microbiome)
- ☐ R / Bioconductor
- ☐ Galaxy
- ☐ Python (Biopython, scikit-bio)
- ☐ Workflow managers (Nextflow, Snakemake)

- ☐ Public databases (NCBI, HOMD, UniProt)
- ☐ None

Other, specify: _____

I05. Faculty with bioinformatics expertise. Number of faculty with bioinformatics or computational-biology expertise

Captures: Human capacity in bioinformatics.

Answer (faculty): _____

I06. Bioinformatics support. Access to bioinformatics support *[domain BIO]*

Captures: Whether biological data can be analysed in-house.

- ☐ (1) None
- ☐ (2) External collaborators only
- ☐ (3) Shared university core facility
- ☐ (4) Dedicated bioinformatician in the department
- ☐ (-1) Don't know / Not applicable

I07. Bioinformatics hours by topic. Hours per term of bioinformatics teaching in this course, by topic (enter 0 if none)

Captures: Time spent teaching bioinformatics, by topic.

Topic / format	Hours (hours per term)
Molecular biology and genetics foundations for omics	
Microbiome analysis	
Sequencing data analysis	
Statistics for high-dimensional data	
Data sharing and FAIR principles	

I08. Bioinformatics teaching depth. Depth of bioinformatics teaching *[domain BIO]*

Captures: Whether students learn to analyse biological data themselves.

- ☐ (1) None
- ☐ (2) Conceptual only
- ☐ (3) Demonstrations
- ☐ (4) Hands-on analysis of real data
- ☐ (-1) Don't know / Not applicable

I09. Omics publications (3 y). Number of peer-reviewed bioinformatics/omics publications by this course in the past 3 years

Captures: Research output in bioinformatics.

Answer (publications): _____

I10. Data sharing (FAIR). Sharing of omics data in public repositories *[domain BIO]*

Captures: Open-science practice, important for consortium-wide reuse of data.

- ☐ (1) Not shared
- ☐ (2) Shared on request
- ☐ (3) Sometimes deposited in public repositories
- ☐ (4) Routinely deposited with FAIR metadata

- ☐ (-1) Don't know / Not applicable

Part J. Governance, ethics, bias and regulation

Goal. Know whether technology is used safely, ethically and under clear responsibility.

What this part captures. Captures the rules around technology use: strategic plans, leadership, oversight, ethics and bias teaching, use of national and professional guidance, patient consent, a governance checklist for AI tools (PAHO AI-GUARD) and stakeholder involvement. Strong governance is what makes technology use safe and trustworthy.

Items J01–J07 refer to the course named in Part A; J08–J12 refer to the policies and procedures that govern it.

J01. Written plan. Does this course have a written plan or formal strategic direction for AI and digital technology in teaching? *[domain GOV]*

Captures: Strategic intent written down.

- ☐ (1) None
- ☐ (2) Informal consensus, not documented
- ☐ (3) Draft document under discussion
- ☐ (4) Formal strategic plan, regularly reviewed
- ☐ (-1) Don't know / Not applicable

J02. Leadership commitment. How committed is the leadership of this course and its department (chair, course director) to advancing these initiatives? *[domain GOV]*

Captures: Leadership support, one of the strongest predictors of change.

- ☐ (1) No commitment shown
- ☐ (2) Initial interest, no concrete actions
- ☐ (3) Some commitment, inconsistent or limited in scope
- ☐ (4) Strong commitment with ongoing initiatives and resources
- ☐ (-1) Don't know / Not applicable

J03. Oversight structure. Is there a committee or a named person accountable for AI and digital technology in teaching and clinics? *[domain GOV]*

Captures: Accountability for technology decisions.

- ☐ (1) None
- ☐ (2) Informal responsibility
- ☐ (3) Committee or person named, limited mandate
- ☐ (4) Formal committee with mandate, reporting line and regular meetings
- ☐ (-1) Don't know / Not applicable

J04. AI ethics teaching. Does this course teach AI ethics (bias, privacy, informed consent, accountability, equity)? *[domain GOV]*

Captures: Preparation of students for ethical use.

- ☐ (1) None
- ☐ (2) Brief mention only
- ☐ (3) 1 dedicated session
- ☐ (4) Systematic integration with case-based discussion and assessment
- ☐ (-1) Don't know / Not applicable

J05. Ethics topics. Which ethics topics are covered?

Captures: The ethics content profile.

- ☐ Algorithmic bias and equity
- ☐ Privacy and data protection
- ☐ Informed consent for technology use with patients
- ☐ Professional accountability / liability
- ☐ Transparency and explainability
- ☐ Automation bias / over-reliance
- ☐ Intellectual property and authorship
- ☐ Environmental impact of AI
- ☐ Academic integrity
- ☐ Genetic data and biobank ethics
- ☐ None

J06. Bias teaching. Are students taught to check AI performance across patient subgroups and to recognize biased training data (PAHO Bias-free AI)? *[domain GOV]*

Captures: A specific skill for fair AI.

- ☐ (1) Not taught
- ☐ (2) Mentioned informally
- ☐ (3) 1 dedicated session
- ☐ (4) Hands-on exercises with real or simulated subgroup analyses
- ☐ (-1) Don't know / Not applicable

J07. Ethical guidelines available. Are established ethical guidelines or white papers on AI and digital technology in dentistry available to faculty and students? *[domain GOV]*

Captures: Access to guidance (PAHO/WHO readiness toolkit item).

- ☐ (1) None available
- ☐ (2) Some documents exist but are incomplete
- ☐ (3) Guidelines exist but need dissemination
- ☐ (4) Comprehensive, well-implemented guidelines available
- ☐ (-1) Don't know / Not applicable

J08. Guidance documents used. Which of the following documents are used in teaching or clinical protocols of this course?

Captures: Which US standards and other guidance documents shape practice.

- ☐ ADA Technical Report 1109, ANSI/ADA Standard 1110-1:2025, ADA White Paper 1106
- ☐ CODA accreditation standards (AI/technology-related elements)
- ☐ ADEA guidance or competencies relating to AI
- ☐ HIPAA privacy and security rules applied to AI tools
- ☐ FDA guidance on AI-enabled device software functions
- ☐ NIST AI Risk Management Framework (AI RMF 1.0) and Generative AI Profile
- ☐ IADR / ITU-WHO Focus Group — Core curriculum for oral and dental AI (Schwendicke et al., 2023)
- ☐ AI reporting guidelines (CONSORT-AI, SPIRIT-AI, TRIPOD+AI, STARD-AI, CLAIM, dental AI checklist)
- ☐ PAHO/WHO AI toolkits (Readiness Assessment, AI-GUARD, Bias-free AI, Prompt design)
- ☐ WHO — Ethics and governance of AI for health (2021) and guidance on large multi-modal models (2024/2025)

- ☐ UNESCO — Guidance for generative AI in education and research (2023)
- ☐ None of these

J09. Regulatory guidance incorporated. Is national / professional regulatory guidance on AI and digital technology in dentistry incorporated into teaching and clinical protocols? *[domain GOV]*

Captures: Translation of regulation into practice.

US reference: see table "US context", row "Professional and regulatory bodies".

- ☐ (1) Not incorporated
- ☐ (2) Aware but not incorporated
- ☐ (3) Discussed in faculty meetings
- ☐ (4) Incorporated into teaching and clinical protocols
- ☐ (-1) Don't know / Not applicable

J10. Patient consent. Are patients informed / asked for consent when AI or robotic systems are used in their care in the school clinic? *[domain GOV]*

Captures: Respect for patient autonomy.

- ☐ (1) Never
- ☐ (2) Occasionally, at clinician discretion
- ☐ (3) Required by policy but inconsistently applied
- ☐ (4) Required by policy, documented in the record
- ☐ (-1) Don't know / Not applicable

J11. AI tool governance (AI-GUARD). Before and during use of clinical AI tools in this course, are the following in place? (adapted from PAHO AI-GUARD, Tier 2)

Captures: A 10-item governance checklist scored Yes = 2, Partial = 1, No = 0 into the AI-GUARD index.

Item	Yes	Partial	No	Don't know
Defined owner / responsible authority for each AI tool	☐	☐	☐	☐
Vendor documentation or model card reviewed (intended use, training data, limitations)	☐	☐	☐	☐
Independent / external validation evidence reviewed	☐	☐	☐	☐
Regulatory clearance verified for the intended use	☐	☐	☐	☐
Performance across patient subgroups evaluated	☐	☐	☐	☐
Human review required before acting on AI output	☐	☐	☐	☐
Clinician override possible and logged	☐	☐	☐	☐
Incident / error reporting process defined	☐	☐	☐	☐
Update / version-change notification procedure defined	☐	☐	☐	☐
Ongoing performance monitoring (drift) planned	☐	☐	☐	☐

Scoring: Yes = 2, Partial = 1, No = 0.

J12. Stakeholder involvement. How involved are stakeholders (faculty, students, clinical staff, IT, patients, external partners) in decisions about technology integration? *[domain GOV]*

Captures: Participation in decisions (PAHO/WHO readiness toolkit item).

- ☐ (1) Not involved
- ☐ (2) Limited, non-systematic involvement
- ☐ (3) Involvement exists but inconsistent
- ☐ (4) Well-established processes for systematic stakeholder input

- ☐ (-1) Don't know / Not applicable

Part K. Implementation, evaluation and sustainability

Goal. Know whether projects move from pilot to routine use, are evaluated, and can last.

What this part captures. Measures whether initiatives move from pilots to routine use and whether they last: pilots, scalability, workflow integration, metrics, evaluation, feedback, cost-benefit, sustainability and funding (PAHO/WHO readiness toolkit). It separates one-off enthusiasm from lasting change.

Refers to technology initiatives of the named course (AI, informatics, robotics, bioinformatics).

K01. Pilot projects. Name any technology pilot project currently running in this course

Captures: Named pilots, their domain and area, to share lessons across schools.

#	Project	Domain	Area
1			
2			
3			

Domain codes: 1 = AI; 2 = Health informatics; 3 = Robotics / digital dentistry; 4 = Bioinformatics

Area codes: 1 = Teaching / learning; 2 = Assessment; 3 = Clinical care; 4 = Research; 5 = Administration

K02. Pilot maturity. Status of technology pilot projects *[domain IMP]*

Captures: Evidence from pilots.

- ☐ (1) No pilot projects
- ☐ (2) Initial pilots with limited outcomes
- ☐ (3) Pilots ongoing, outcomes not yet assessed
- ☐ (4) Pilots with positive, documented outcomes
- ☐ (-1) Don't know / Not applicable

K03. Scalability. Potential for scaling successful initiatives to other courses *[domain IMP]*

Captures: Whether success can spread.

- ☐ (1) No potential
- ☐ (2) Limited potential; substantial effort required
- ☐ (3) Some potential, requires significant resources and coordination
- ☐ (4) High potential with a clear pathway
- ☐ (-1) Don't know / Not applicable

K04. Workflow integration. How well are these technologies integrated into clinical and teaching workflows? *[domain IMP]*

Captures: Routine use versus side projects.

- ☐ (1) Not integrated
- ☐ (2) Used outside routine workflow
- ☐ (3) Partly integrated
- ☐ (4) Fully integrated in routine workflow
- ☐ (-1) Don't know / Not applicable

K05. Performance metrics. Are there established metrics for monitoring the performance or impact of technology teaching initiatives? *[domain IMP]*

Captures: Measurement of results.

- ☐ (1) No metrics
- ☐ (2) Limited metrics
- ☐ (3) Metrics exist but need refinement
- ☐ (4) Comprehensive metrics with regular review
- ☐ (-1) Don't know / Not applicable

K06. Evaluation framework. Is the impact of technology teaching evaluated with a framework (e.g., Kirkpatrick levels: reaction, learning, behaviour, results)? *[domain IMP]*

Captures: Depth of evaluation.

- ☐ (1) Not evaluated
- ☐ (2) Student satisfaction only
- ☐ (3) Learning outcomes measured
- ☐ (4) Learning and clinical/behavioural outcomes measured
- ☐ (-1) Don't know / Not applicable

K07. Feedback mechanisms. Are there mechanisms to collect and act on feedback from students and faculty? *[domain IMP]*

Captures: Listening to users.

- ☐ (1) None
- ☐ (2) Informal feedback only
- ☐ (3) Formal collection, inconsistent follow-up
- ☐ (4) Formal collection with documented actions
- ☐ (-1) Don't know / Not applicable

K08. Continuous improvement. Are there mechanisms for continuous improvement based on evaluation? *[domain IMP]*

Captures: Learning from evaluation.

- ☐ (1) No mechanisms
- ☐ (2) Limited mechanisms
- ☐ (3) Mechanisms exist but need better integration
- ☐ (4) Comprehensive feedback loops
- ☐ (-1) Don't know / Not applicable

K09. Cost-benefit evaluation. Has the cost and benefit of technology initiatives been evaluated (licences, hardware, staff time vs. outcomes)? *[domain IMP]*

Captures: Economic evidence for investment decisions.

- ☐ (1) Never
- ☐ (2) Informal estimate
- ☐ (3) Formal evaluation for some initiatives
- ☐ (4) Formal evaluation for all major initiatives
- ☐ (-1) Don't know / Not applicable

K10. Financial sustainability. Are there plans for long-term financial sustainability (budget lines, grant strategies, partnerships)? *[domain IMP]*

Captures: Whether initiatives will survive after initial funding.

- ☐ (1) No plans
- ☐ (2) Initial discussions only
- ☐ (3) Plans exist but incomplete
- ☐ (4) Comprehensive, well-developed plans
- ☐ (-1) Don't know / Not applicable

K11. Funding sources. Name each funding source that supports technology teaching or research in this course

Captures: Who pays, to identify funding opportunities for consortium projects.

#	Funding source	Type
1		
2		
3		

Type codes: 1 = Institutional budget; 2 = National research agency (e.g., NIH, CIHR, CNPq/CAPES/FAPs, NSFC, ICMR/DST); 3 = Industry; 4 = Philanthropy / foundation; 5 = International agency; 6 = Other

K12. Funding adequacy. Funding available for technology teaching or research in this course *[domain IMP]*

Captures: Adequacy of money.

- ☐ (1) No funding
- ☐ (2) Limited funding with significant gaps
- ☐ (3) Funding available but inconsistent or insufficient
- ☐ (4) Sufficient, reliable funding
- ☐ (-1) Don't know / Not applicable

K13. Annual budget. Approximate annual budget dedicated to these technologies in this course (USD equivalent)

Captures: Order of magnitude of investment, comparable across schools.

- ☐ None
- ☐ Less than 5,000
- ☐ 5,000 – 25,000
- ☐ 25,001 – 100,000
- ☐ More than 100,000
- ☐ Don't know

K14. Grants (3 y). Number of competitive grants related to these technologies obtained by this course in the past 3 years

Captures: Success in attracting competitive funding.

Answer (grants): _____

Part L. Collaboration and the open dental AI model

Goal. Know whether the course is ready to share data and work with other institutions on AI.

What this part captures. Maps which departments could contribute data, annotation, validation, teaching or grant work to the consortium's free dental AI model, under which conditions, and what stands in the way. It turns the survey into a practical plan for collaboration.

These items measure readiness to collaborate and share data in multi-institution AI projects, such as an open dental AI model. Answering does not commit your institution.

L01. Interest in multi-institution projects. Interest of this course in joining multi-institution AI projects (for example, ICARB-D projects)

Captures: Who wants to join, to plan the first collaborative projects.

- ☐ Not interested
- ☐ Would like more information
- ☐ Interested in participating
- ☐ Already involved

L02. Data that could be contributed. Which de-identified data could this course potentially contribute to a shared dental AI model (subject to ethics approval)?

Captures: The potential data pool for the open AI model, by modality.

- ☐ Periapical / bitewing radiographs
- ☐ Panoramic radiographs
- ☐ CBCT
- ☐ Intraoral scans (3D)
- ☐ Clinical photographs
- ☐ Structured clinical records
- ☐ Histopathology images
- ☐ Omics / microbiome data
- ☐ None

L03. Images available for sharing. Approximate number of de-identified images or records that could be shared (0 if none)

Captures: Scale of potential contribution.

Answer (images/records): _____

L04. Ability to share data. Ability to share de-identified data with external partners

Captures: How quickly data could actually be shared (not part of the maturity score).

- ☐ (1) Not possible
- ☐ (2) Possible only with a new ethics approval
- ☐ (3) Possible under existing approvals and data-sharing agreements
- ☐ (4) Already sharing data with external partners
- ☐ (-1) Don't know / Not applicable

L05. Preferred ways to collaborate. How would this course prefer to collaborate?

Captures: The collaboration models each partner accepts, e.g., federated learning when data cannot leave the institution.

- ☐ Federated learning (data stays at our institution)
- ☐ Contributing de-identified data to a central repository
- ☐ Annotating / labelling images
- ☐ Testing and validating models on our own data
- ☐ Sharing curricula and teaching materials
- ☐ Joint grant applications and publications

Other, specify: _____

L06. Barriers to sharing. Main barriers to sharing data or collaborating

Captures: Obstacles the consortium must solve (e.g., model agreements, federated tools).

- □ Privacy law / legal uncertainty
- □ Ethics approval
- □ Institutional policy
- □ Technical capacity
- □ Data quality or lack of annotation
- □ Intellectual property concerns
- □ Staff time
- □ No major barriers

Other, specify: _____

L07. People available. Number of faculty, staff or graduate students who could dedicate time to consortium projects

Captures: Human capacity available for joint work.

Answer (people): _____

L08. Project ideas. Ideas for collaborative projects (optional)

Captures: Suggestions from the field to shape the consortium agenda.

Part M. Self-assessment and open response

Goal. Compare how the course sees itself with what its answers show, and record its most urgent needs.

What this part captures. Asks the respondent for an overall judgement of maturity in health informatics as a whole and in each of its areas, the main barriers and facilitators, and open comments. Self-ratings are compared with the scores computed from Parts C–K, and open answers explain the numbers.

M01. Overall maturity — AI. Overall maturity of this course in artificial intelligence

Captures: Respondent's overall judgement for AI, compared with computed scores.

- □ (1) Exploring: no formal action; individual interest only
- □ (2) Planning: discussions underway, some resources identified, no systematic implementation
- □ (3) Implementing: some formal courses/projects launched, not scaled or sustainable
- □ (4) Mature: systematic integration, dedicated resources, continuous evaluation
- □ (-1) Don't know / Not applicable

M02. Overall maturity — health information systems. Overall maturity in health information systems (records, imaging, coding, teledentistry)

Captures: Overall judgement for health information systems.

- □ (1) Exploring: no formal action; individual interest only
- □ (2) Planning: discussions underway, some resources identified, no systematic implementation
- □ (3) Implementing: some formal courses/projects launched, not scaled or sustainable
- □ (4) Mature: systematic integration, dedicated resources, continuous evaluation
- □ (-1) Don't know / Not applicable

M03. Overall maturity — robotics. Overall maturity in robotics and digital dentistry

Captures: Overall judgement for robotics and digital dentistry.

- ☐ (1) Exploring: no formal action; individual interest only
- ☐ (2) Planning: discussions underway, some resources identified, no systematic implementation
- ☐ (3) Implementing: some formal courses/projects launched, not scaled or sustainable
- ☐ (4) Mature: systematic integration, dedicated resources, continuous evaluation
- ☐ (-1) Don't know / Not applicable

M04. Overall maturity — bioinformatics. Overall maturity in bioinformatics and omics

Captures: Overall judgement for bioinformatics.

- ☐ (1) Exploring: no formal action; individual interest only
- ☐ (2) Planning: discussions underway, some resources identified, no systematic implementation
- ☐ (3) Implementing: some formal courses/projects launched, not scaled or sustainable
- ☐ (4) Mature: systematic integration, dedicated resources, continuous evaluation
- ☐ (-1) Don't know / Not applicable

M05. Primary barriers. Primary barriers to progress in this course (select up to three)

Captures: The three biggest obstacles, ranked across schools.

Select up to 3.

- ☐ Lack of funding
- ☐ Lack of faculty expertise
- ☐ Lack of time in the curriculum
- ☐ Lack of leadership support
- ☐ Lack of equipment / infrastructure
- ☐ Ethical / legal uncertainty
- ☐ Lack of student interest or preparedness
- ☐ Lack of data access
- ☐ Accreditation / curriculum rules leave no room
- ☐ Language barriers (tools/materials not in local language)

Other, specify: _____

M06. Facilitators. Main facilitators of progress (select up to three)

Captures: What helps, to replicate success elsewhere.

Select up to 3.

- ☐ Faculty champions
- ☐ Leadership support
- ☐ Dedicated funding
- ☐ Interdisciplinary partners
- ☐ Industry partnerships
- ☐ Student demand
- ☐ Accreditation / national guidelines
- ☐ National or international networks

Other, specify: _____

M07. Most important need. What is the single most important resource or policy support this course needs in the next 2 years?

Captures: The top priority in the respondent's words, analysed by theme.

M08. Innovative practice. Describe any innovative teaching practice with AI, informatics, robotics or bioinformatics in this course (optional)

Captures: Good practices to showcase and share.

M09. Vision 2030. How do you expect these technologies to change teaching in this course by 2030? (optional)

Captures: Expectations for the future (PAHO "Vision 2030" question).

M10. Confidence in answers. How confident are you in the accuracy of your answers?

Captures: Data-quality flag used in sensitivity analyses.

- ☐ (1) Low
- ☐ (2) Moderate
- ☐ (3) High
- ☐ (4) Very high (checked with records)

M12. Overall maturity — software development. Overall maturity in software development for health

Captures: Overall judgement for software development for health.

- ☐ (1) Exploring: no formal action; individual interest only
- ☐ (2) Planning: discussions underway, some resources identified, no systematic implementation
- ☐ (3) Implementing: some formal courses/projects launched, not scaled or sustainable
- ☐ (4) Mature: systematic integration, dedicated resources, continuous evaluation
- ☐ (-1) Don't know / Not applicable

M13. Overall maturity — health informatics. Overall maturity in health informatics as a whole (all of the areas above)

Captures: Overall judgement for the umbrella field, compared with the benchmark score.

- ☐ (1) Exploring: no formal action; individual interest only
- ☐ (2) Planning: discussions underway, some resources identified, no systematic implementation
- ☐ (3) Implementing: some formal courses/projects launched, not scaled or sustainable
- ☐ (4) Mature: systematic integration, dedicated resources, continuous evaluation
- ☐ (-1) Don't know / Not applicable

M11. Follow-up interview. Would you agree to a short follow-up interview?

Captures: Willingness for qualitative follow-up.

- ☐ Yes
- ☐ No

3. Scoring and consolidation

- **Course record.** Each form is stored as one record keyed by Survey year × Country × State × School × Department × Course code. Nothing is aggregated before storage.
- **Domain scores.** For each record, the mean of the 1–4 items of each of the eleven domains is computed ("Don't know" = missing) and converted to a 0–100 index: (mean − 1) / 3 × 100. The overall index is the mean of the domain indices.
- **AI-GUARD and safeguard indices.** Items J11 and F06 are scored Yes = 2, Partial = 1, No = 0 and expressed as a percentage of the maximum over answered rows.
- **Hours.** Hours per year = hours per term × terms per year (B05), for AI (E02, E03, E09), health informatics (G09), robotics (H05) and bioinformatics (I07).
- **Equipment.** Equipment counts (H01) are summed and divided by enrollment (B01) to give students per unit.
- **Department roll-up.** Scores are averaged across the courses of a department in the same year; the number of contributing courses (n) is reported.
- **School roll-up.** Department scores are averaged with weights equal to the number of courses each department contributed.
- **Benchmark.** The benchmark of a course, department or school is the mean of its eleven domain indices (0–100); for a school it is built from its departments, weighted by courses. Schools can be listed by benchmark within each year, with the number of departments and courses shown.
- **Readiness to use new funding.** The mean of the five foundation domains (INF, DAT, HUM, GOV, IMP) indicates which schools are most likely to use a grant for health informatics well.
- **State, national and country summaries.** School scores are averaged per state, per country and per year; the number of schools is reported.
- **Annual trends.** The overall index of each department is tabulated by survey year to measure progress between waves.
- **Transparency rule.** For each school, the list of named courses that produced the curriculum score (Part E) is published.

Data file

Answers are stored in a flat file with one row per department/course (access code) and one column per item — 469 answer columns in total. The codebook (docs/codebook.csv) lists every column, its label, type, domain and value codes. Personal contact data (respondent name, email, interviewer) are kept in a separate private file.

Methodological recommendations

- Content validity: expert panel (at least two experts per domain) rating item relevance; item-level and scale-level content validity index (I-CVI ≥ 0.78, S-CVI/Ave ≥ 0.90), optionally with a two-round Delphi.
- Pilot: cognitive interviews with 2–3 department heads, then a pilot with about 10 departments to check timing and item clarity.
- Reliability: Cronbach's alpha / McDonald's omega per domain; test–retest on a 10% subsample after 2–4 weeks; agreement between two respondents of the same department where possible.
- Sampling frame: the CODA list of accredited US dental schools and ADEA member institutions. Stratify by public/private status and US region; within each school, invite departments teaching imaging, restorative/prosthodontics, orthodontics, periodontics, oral surgery/implantology, oral biology, public health and research methods, plus any department teaching a dedicated technology course.
- Annual design: the same departments are invited every year (panel), new US schools join as the consortium grows. Keep item IDs stable between waves; record any change in a versioned codebook.

- Ethics: approval from the coordinating research ethics board and local institutional review board (IRB) approval where required. Respondents provide informed consent (item A14). Personal contact data are stored separately from answers.
- Reporting: STROBE for cross-sectional studies and CHERRIES for web-based surveys.
- Analysis: descriptive statistics by state, region, level and year; Kruskal–Wallis / Mann–Whitney tests for regional and public/private comparisons; multilevel (mixed-effects) models with courses nested in departments and schools, and survey year as a repeated measure; heat maps of domain indices.

4. Maturity checklist

After answering, the same items work as a guide. Each of the 91 points below comes from a 1–4 item of the survey (or a row of an AI-GUARD checklist). The target is level 4 (established); the next step is level 3 (developing). A school, department or course can tick what is already true and plan the steps that lead to the level it wants to reach.

Level	Meaning
1 · Absent	Nothing in place yet.
2 · Initial	Interest or informal, isolated efforts.
3 · Developing	In place in part, or not yet routine.
4 · Established	Routine, documented and supported.

Part C. Physical resources: computing, AI tools and data

Goal. Know what the course has today: computing, software, AI tools, data, privacy and security.

- ☐ **Computing hardware** (C01)
  Target: Dedicated high-performance computing, regularly updated and accessible to students
  Next step: Shared equipment, insufficient for demand
- ☐ **Cloud computing** (C03)
  Target: Routine institutional access with support
  Next step: Institutional access on request, limited quota
- ☐ **Dedicated digital space** (C05)
  Target: Dedicated AI/digital dentistry lab with staff and scheduled student access
  Next step: Dedicated digital dentistry lab without AI focus
- ☐ **Number of AI tools** (C09)
  Target: 4 or more AI tools used and integrated into graded coursework
  Next step: 2–3 AI tools used
- ☐ **De-identified dataset** (C10)
  Target: Dataset exists with clear access protocols and is used in the course
  Next step: Dataset exists but access is restricted and scale is limited
- ☐ **Annotation capacity** (C13)
  Target: Calibrated annotators with documented protocol and agreement statistics
  Next step: Annotation protocol exists, limited calibration
- ☐ **Data quality** (C14)
  Target: Systematic quality assurance with documented protocols
  Next step: Some measures exist but are not comprehensive
- ☐ **Data representativeness** (C15)
  Target: Systematically assessed and reported
  Next step: Assessed for some projects
- ☐ **Privacy and security** (C16)
  Target: Comprehensive systems with encryption, audit trails and a named person responsible
  Next step: Systems exist but are inconsistent
- ☐ **De-identification** (C17)
  Target: Written procedures, routinely applied and audited
  Next step: Written procedures, inconsistently applied
- ☐ **Secure research environment** (C18)
  Target: Available and routinely used
  Next step: Available on request with delays
- ☐ **Connectivity and IT support** (C19)
  Target: Dedicated educational-technology support assigned to the unit
  Next step: Partial support (software installation only)

- ☐ **Interoperability** (C20)
  Target: Well-established standards enabling seamless integration
  Next step: Standards exist but implementation is inconsistent
- ☐ **Cybersecurity** (C21)
  Target: Specific security review before adoption and periodic audit
  Next step: Measures exist but not specific to these tools
- ☐ **Equitable student access** (C22)
  Target: Equal access during and outside class hours
  Next step: Access during scheduled sessions only

Part D. Human resources and expertise (AI and data science)

Goal. Know who can teach, build and support these technologies, and what holds them back.

- ☐ **Faculty able to teach AI** (D01)
  Target: More than 25%
  Next step: 11–25%
- ☐ **AI-primary researchers** (D04)
  Target: 4 or more
  Next step: 2–3
- ☐ **Faculty AI training** (D07)
  Target: Systematic training with institutional incentives and tracking
  Next step: Organized training, but low or inconsistent attendance
- ☐ **Partnership depth** (D11)
  Target: Long-term institutional partnership with joint funding or appointments
  Next step: Formal project-based collaboration
- ☐ **Technical staff** (D12)
  Target: Dedicated technical staff embedded in the department
  Next step: Adequate access for most needs
- ☐ **Hiring plans** (D14)
  Target: Expertise already recruited and further positions planned
  Next step: Position approved or recruitment underway
- ☐ **Faculty openness** (D15)
  Target: Strongly supportive with active champions
  Next step: Mostly supportive

Part E. AI curriculum and teaching

Goal. Know what students actually learn about AI, at which level and for how many hours.

- ☐ **Dedicated AI instruction** (E01)
  Target: 1 or more dedicated AI courses, or systematic integration across sessions
  Next step: 1 dedicated module or class session (≤4 hours)
- ☐ **Technical depth** (E06)
  Target: Includes programming or formal AI development
  Next step: Hands-on operation of existing AI tools under guidance
- ☐ **Programming / AI development** (E07)
  Target: Students independently complete a small AI project (e.g., train and validate a classifier)
  Next step: Hands-on exercises under guidance
- ☐ **Graduate AI offering** (E13)
  Target: Required course plus a formal AI research project requirement
  Next step: Required course exists
- ☐ **Clinical AI training** (E15)
  Target: Integrated into clinical rotations with formal assessment
  Next step: 1 dedicated session
- ☐ **Assessment of AI competency** (E16)
  Target: Competency-based assessment (OSCE station, portfolio, practical demonstration)

Next step: Separate AI assignment or project

- ☐ **Educational materials** (E18)
  Target: Comprehensive, curated collection integrated into the course
  Next step: Some dedicated materials used
- ☐ **Competency framework** (E20)
  Target: Explicitly mapped to a framework with defined learning outcomes
  Next step: Partially aligned
- ☐ **AI as a teaching tool** (E21)
  Target: Systematically used and evaluated
  Next step: Used regularly in some activities
- ☐ **Student preparedness** (E23)
  Target: High interest and preparedness
  Next step: Interest and moderate preparedness
- ☐ **Planned curriculum change** (E24)
  Target: Being implemented now
  Next step: Approved, not yet implemented

Part F. Generative AI in teaching and learning

Goal. Know whether generative AI is used with clear rules, real skills and working safeguards.

- ☐ **Generative AI policy** (F01)
  Target: Formal, regularly updated policy with training
  Next step: Interim guidance issued
- ☐ **Prompt design teaching** (F04)
  Target: Integrated practice with feedback
  Next step: 1 dedicated session
- ☐ **Verification of outputs** (F05)
  Target: Integrated practice with assessment
  Next step: 1 dedicated session
- ☐ **Rule that no identifiable patient data is entered into public AI tools** (F06_no_patient_data)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **An institutionally licensed, privacy-protected generative AI tool is available** (F06_secure_tool)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Students must disclose AI use in assignments** (F06_disclosure)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Human verification required before AI output is used in patient care** (F06_human_review)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Process to report misuse or AI errors** (F06_incident)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Staff and students trained on limitations and bias of generative AI** (F06_training)
  Target: Yes: fully in place
  Next step: Partial

Part G. Health information systems and software development

Goal. Know whether the clinic produces reliable digital data, and whether the course can build and maintain its own health software.

- ☐ **Electronic health record** (G01)
  Target: Integrated electronic record with structured data, imaging and decision support
  Next step: Electronic clinical records, mostly free text

- ☐ **Digital imaging archive** (G04)
  Target: Enterprise PACS integrated with the health record via DICOM
  Next step: Central archive (PACS) for some modalities
- ☐ **Teledentistry** (G05)
  Target: Routine teledentistry with student involvement
  Next step: Structured program in some services
- ☐ **Data analytics** (G06)
  Target: Routine dashboards used for clinical and teaching quality decisions
  Next step: Regular reports for some indicators
- ☐ **Research access to clinical data** (G07)
  Target: Curated research data warehouse with governance
  Next step: Formal request process
- ☐ **Public-health informatics / GIS** (G08)
  Target: Routinely
  Next step: In some projects or courses
- ☐ **Informatics teaching depth** (G10)
  Target: Dedicated course with practical assessment
  Next step: Dedicated module
- ☐ **Clinical informatics staff** (G12)
  Target: Dedicated clinical informatics team for the school
  Next step: Shared clinical informatics staff
- ☐ **Software development capacity** (G13)
  Target: Software built and maintained by a team, with documentation and real users
  Next step: Some tools built for teaching or research, not maintained
- ☐ **Development practices** (G14)
  Target: Version control, testing, code review, and security and privacy by design
  Next step: Version control and some testing
- ☐ **Teaching software development** (G15)
  Target: Structured course or project-based learning with real users
  Next step: One dedicated session or module

Part H. Robotics and digital dentistry

Goal. Know what robotic and digital equipment exists, and whether students really use it.

- ☐ **Student access to robotics** (H03)
  Target: Regular hands-on practice including supervised clinical cases
  Next step: Supervised hands-on practice in preclinic
- ☐ **Digital workflow** (H04)
  Target: Full digital workflow routine across clinics and teaching
  Next step: Full digital workflow in some clinics
- ☐ **Maintenance and replacement** (H08)
  Target: Full maintenance and replacement plan with budget
  Next step: Service contracts for some equipment
- ☐ **Safety and credentialing** (H09)
  Target: Formal credentialing with documented training and supervision
  Next step: Written protocol, not always applied
- ☐ **Robotics research** (H10)
  Target: Established research group with engineering partners
  Next step: Funded projects

Part I. Bioinformatics and omics

Goal. Know whether the course can produce, analyse and share biological data.

- ☐ **Biobank and sample governance** (I02)
  Target: Biobank with linked clinical data, governance and use by several projects

Next step: Biobank with consent protocol

- ☐ **Bioinformatics support** (I06)
  Target: Dedicated bioinformatician in the department
  Next step: Shared university core facility
- ☐ **Bioinformatics teaching depth** (I08)
  Target: Hands-on analysis of real data
  Next step: Demonstrations
- ☐ **Data sharing (FAIR)** (I10)
  Target: Routinely deposited with FAIR metadata
  Next step: Sometimes deposited in public repositories

Part J. Governance, ethics, bias and regulation

Goal. Know whether technology is used safely, ethically and under clear responsibility.

- ☐ **Written plan** (J01)
  Target: Formal strategic plan, regularly reviewed
  Next step: Draft document under discussion
- ☐ **Leadership commitment** (J02)
  Target: Strong commitment with ongoing initiatives and resources
  Next step: Some commitment, inconsistent or limited in scope
- ☐ **Oversight structure** (J03)
  Target: Formal committee with mandate, reporting line and regular meetings
  Next step: Committee or person named, limited mandate
- ☐ **AI ethics teaching** (J04)
  Target: Systematic integration with case-based discussion and assessment
  Next step: 1 dedicated session
- ☐ **Bias teaching** (J06)
  Target: Hands-on exercises with real or simulated subgroup analyses
  Next step: 1 dedicated session
- ☐ **Ethical guidelines available** (J07)
  Target: Comprehensive, well-implemented guidelines available
  Next step: Guidelines exist but need dissemination
- ☐ **Regulatory guidance incorporated** (J09)
  Target: Incorporated into teaching and clinical protocols
  Next step: Discussed in faculty meetings
- ☐ **Patient consent** (J10)
  Target: Required by policy, documented in the record
  Next step: Required by policy but inconsistently applied
- ☐ **Defined owner / responsible authority for each AI tool** (J11_ownership)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Vendor documentation or model card reviewed (intended use, training data, limitations)** (J11_vendor_docs)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Independent / external validation evidence reviewed** (J11_validation)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Regulatory clearance verified for the intended use** (J11_regulatory)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Performance across patient subgroups evaluated** (J11_subgroup)
  Target: Yes: fully in place
  Next step: Partial

- ☐ **Human review required before acting on AI output** (J11_human_review)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Clinician override possible and logged** (J11_override)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Incident / error reporting process defined** (J11_incident)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Update / version-change notification procedure defined** (J11_updates)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Ongoing performance monitoring (drift) planned** (J11_monitoring)
  Target: Yes: fully in place
  Next step: Partial
- ☐ **Stakeholder involvement** (J12)
  Target: Well-established processes for systematic stakeholder input
  Next step: Involvement exists but inconsistent

Part K. Implementation, evaluation and sustainability

Goal. Know whether projects move from pilot to routine use, are evaluated, and can last.

- ☐ **Pilot maturity** (K02)
  Target: Pilots with positive, documented outcomes
  Next step: Pilots ongoing, outcomes not yet assessed
- ☐ **Scalability** (K03)
  Target: High potential with a clear pathway
  Next step: Some potential, requires significant resources and coordination
- ☐ **Workflow integration** (K04)
  Target: Fully integrated in routine workflow
  Next step: Partly integrated
- ☐ **Performance metrics** (K05)
  Target: Comprehensive metrics with regular review
  Next step: Metrics exist but need refinement
- ☐ **Evaluation framework** (K06)
  Target: Learning and clinical/behavioural outcomes measured
  Next step: Learning outcomes measured
- ☐ **Feedback mechanisms** (K07)
  Target: Formal collection with documented actions
  Next step: Formal collection, inconsistent follow-up
- ☐ **Continuous improvement** (K08)
  Target: Comprehensive feedback loops
  Next step: Mechanisms exist but need better integration
- ☐ **Cost-benefit evaluation** (K09)
  Target: Formal evaluation for all major initiatives
  Next step: Formal evaluation for some initiatives
- ☐ **Financial sustainability** (K10)
  Target: Comprehensive, well-developed plans
  Next step: Plans exist but incomplete
- ☐ **Funding adequacy** (K12)
  Target: Sufficient, reliable funding
  Next step: Funding available but inconsistent or insufficient

5. Related studies

Most published studies on AI in dental education measure perceptions, knowledge and attitudes of students or educators, often at one institution, and health informatics, robotics and bioinformatics are rarely surveyed at all. No national study has measured the digital maturity of US dental schools at the level of named departments and courses, combining physical and human resources, teaching hours by topic, governance and sustainability, and repeating the measurement every year. This survey fills that gap for the United States.

Study	Setting / design	Key finding
PAHO/WHO & IDB, 2024	Readiness toolkit	Defines the readiness dimensions and the scoring/analysis steps adapted here.
PAHO/WHO & IDB, 2026 (AI-GUARD)	Governance tool	Tiered Yes/Partial/No governance checklist for AI tools; basis of items F06 and J11.
PAHO, 2026 (Bias-free AI)	Guidance	Representative data, subgroup performance reporting and human oversight; basis of items C15 and J06.
PAHO, 2025 (Prompt design)	Guidance	Prompt design as a core digital-literacy competency; basis of items F04–F05.
Schwendicke et al., 2023	Expert consensus (IADR, ITU/WHO FG-AI4H), Delphi	Core curriculum for dental AI in four domains, mostly at "knowledge" level; reference framework for item E20.
Uribe et al., 2024	Worldwide survey of 428 dental educators	31% already used AI chatbots, 64% saw potential; main concern was the lack of guidelines and training.
Kavadella et al., 2026	Multinational survey of dental students	Students want AI education (about 70%) but over half feel unprepared by current curricula.
El-Hakim et al., 2025	Scoping review, 17 studies	AI improves procedural accuracy and feedback but struggles with dexterity and judgment; VR often mislabelled as AI.
Sheng et al., 2026	243 faculty, residents, students; UTHealth Houston	66% of faculty and 73% of students use LLM tools; faculty report greater need for AI training.
Busch et al., 2024	4,596 medical, dental, veterinary students; 192 faculties, 48 countries	Positive attitudes and demand for more AI teaching; only 205 dental students.
Bisdas et al., 2021	3,133 medical and dental students, 63 countries	72% see AI as a partner rather than a competitor.
NIDCR, 2022	Federal research strategy	Calls for health informatics and data-science research and for training programs that use data science.
NIH, 2021	National report	AI, 3D printing and digital imaging are reshaping oral health care; about 60 million Americans live in areas with too few providers.

Suggested article titles

- A Benchmark of Health Informatics Maturity in US Dental Schools: A National, Annual, Course-Level Survey of Artificial Intelligence, Robotics, Bioinformatics and Software Development
- Which Dental School Is Ready to Invest? Physical and Human Resources for Health Informatics in US Dental Schools — Baseline Results of the ICARB-D Annual Survey
- From Courses to Universities: Measuring Health Informatics Maturity in US Dental Education with an Adapted PAHO/WHO Readiness Framework

Candidate journals

Journal of Dental Education (ADEA) · Journal of the American Dental Association (JADA) · JDR Clinical & Translational Research · BMC Medical Education · JMIR Medical Education · Frontiers in Dental Medicine.

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